

TBF Condition Monitoring - AI-Powered Predictive Maintenance Dashboard

Yeh project Tunnel Booster Fans (TBFs) ke liye ek AI-based predictive maintenance system ka prototype hai. Yeh simulated sensor data use karke RUL (Remaining Useful Life), Fault Diagnosis, aur Anomaly Detection karta hai.

****Developed By:**** Rajeev Mandle, Ayush Mark Ekka, Janmay Jai Singh (Bhilai Institute of Technology Durg)

File Structure (Folder Layout)

Project ko run karne ke liye, aapke folder ka structure aisa hona chahiye:

Your_Project_Folder_Name/

- |— assets/ # CSS aur images ke liye folder
 - |— style.css # UI styling ke liye file
- |— pages/ # Dash multi-page feature ke liye folder
 - |— landing.py # Landing page ka code
 - |— dashboard.py # Main dashboard ka code
- |— config.py # Saari configuration settings
- |— simulation.py # TBF simulation logic (TBFState class)
- |— data_processing.py # Data generation aur feature engineering functions
- |— ml_models.py # Model training/loading aur prediction functions
- |— shared_state.py # Shared variables/logger (GLOBAL_STATE)
- |— dash_app.py # Main Dash application file (isko run karna hai)
- |— requirements.txt # Zaroori Python libraries ki list
- |— tbf_historical_data.csv (Optional - agar nahi hai toh generate ho jayegi)
- |— tbf_rul_model.pkl (Optional - agar nahi hai toh train ho jayega)
- |— tbf_fault_model.pkl (Optional - agar nahi hai toh train ho jayega)
- |— tbf_anomaly_model.pkl (Optional - agar nahi hai toh train ho jayega)
- |— tbf_anomaly_scaler.pkl (Optional - agar nahi hai toh train ho jayega)
- |— tbf_monitor_streamlit.log (Log file - automatically banegi)

Prerequisites (Pehle Se Kya Chahiye)

* **Python:** Version 3.8 ya usse naya install hona chahiye. (Humne 3.13 use kiya tha, par 3.8+ chalna chahiye). Aap [python.org](https://www.python.org/downloads/) se download kar sakte hain.

* **pip:** Python ka package installer (usually Python ke saath aata hai).

Setup Instructions (Kaise Install Karein)

1. **Get the Code:** Project ki saari files (upar diye gaye structure ke hisaab se) apne dost ke laptop pe copy karo. Ya agar Git use kar rahe ho toh clone karo.

2. **Open Terminal:** Us folder mein jao jahan aapne project copy kiya hai (Command Prompt, PowerShell, Git Bash, etc. use karke).

```
```bash
cd path/to/Your_Project_Folder_Name
```
```

3. **Create Virtual Environment (Recommended):** Ek virtual environment banana acchi practice hai taaki project ki libraries system ki libraries se mix na hon.

```
```bash
Create virtual environment (venv naam ka folder banega)
python -m venv venv
Activate virtual environment
Windows pe:
.\venv\Scripts\activate
MacOS/Linux pe:
source venv/bin/activate
```
```

(Jab kaam ho jaaye, deactivate karne ke liye sirf `deactivate` type karna hota hai).

4. **Install Dependencies:** Ab zaroori libraries install karo `requirements.txt` file use karke.

```
```bash
pip install -r requirements.txt
```
```

(Yeh command neeche di gayi saari libraries install kar degi).

Dependencies (Kya Install Hoga)

`requirements.txt` file mein yeh libraries honi chahiye:

```
``text
pandas
numpy
scikit-learn
joblib
plotly
dash >= 2.10
dash-bootstrap-components >= 1.4
```

Running the Application (Kaise Chalayein)

1. Ensure karo ki aapka virtual environment activated hai (agar banaya tha toh) aur aap project folder mein ho terminal ke andar.
2. Neeche di gayi command run karo:
`python dash_app.py`
3. Terminal mein output aayega, kuch aisa:
Dash is running on http://127.0.0.1:8050/
* Serving Flask app 'dash_app'
* Debug mode: on
... (baaki logs) ...
4. Apne web browser (Chrome, Firefox, etc.) mein yeh URL open karo:
`http://127.0.0.1:8050/`
5. App ka landing page khul jaayega. Wahan se "Launch Dashboard" button click karke monitoring dashboard pe jaa sakte ho.

Notes

- Agar .pkl model files ya tbf_historical_data.csv file project folder mein nahi hongi, toh app pehli baar start hone par unhe automatically generate/train karne ki koshish karega. Is